

Chapter 14

Pulse-Width Modulator (PWM8B8CV1)

14.1 Introduction

The PWM definition is based on the HC12 PWM definitions. It contains the basic features from the HC11 with some of the enhancements incorporated on the HC12: center aligned output mode and four available clock sources. The PWM module has eight channels with independent control of left and center aligned outputs on each channel.

Each of the eight channels has a programmable period and duty cycle as well as a dedicated counter. A flexible clock select scheme allows a total of four different clock sources to be used with the counters. Each of the modulators can create independent continuous waveforms with software-selectable duty rates from 0% to 100%. The PWM outputs can be programmed as left aligned outputs or center aligned outputs.

14.1.1 Features

The PWM block includes these distinctive features:

- Eight independent PWM channels with programmable period and duty cycle
- Dedicated counter for each PWM channel
- Programmable PWM enable/disable for each channel
- Software selection of PWM duty pulse polarity for each channel
- Period and duty cycle are double buffered. Change takes effect when the end of the effective period is reached (PWM counter reaches zero) or when the channel is disabled.
- Programmable center or left aligned outputs on individual channels
- Eight 8-bit channel or four 16-bit channel PWM resolution
- Four clock sources (A, B, SA, and SB) provide for a wide range of frequencies
- Programmable clock select logic
- Emergency shutdown

14.1.2 Modes of Operation

There is a software programmable option for low power consumption in wait mode that disables the input clock to the prescaler.

In freeze mode there is a software programmable option to disable the input clock to the prescaler. This is useful for emulation.

14.1.3 Block Diagram

Figure 14-1 shows the block diagram for the 8-bit 8-channel PWM block.

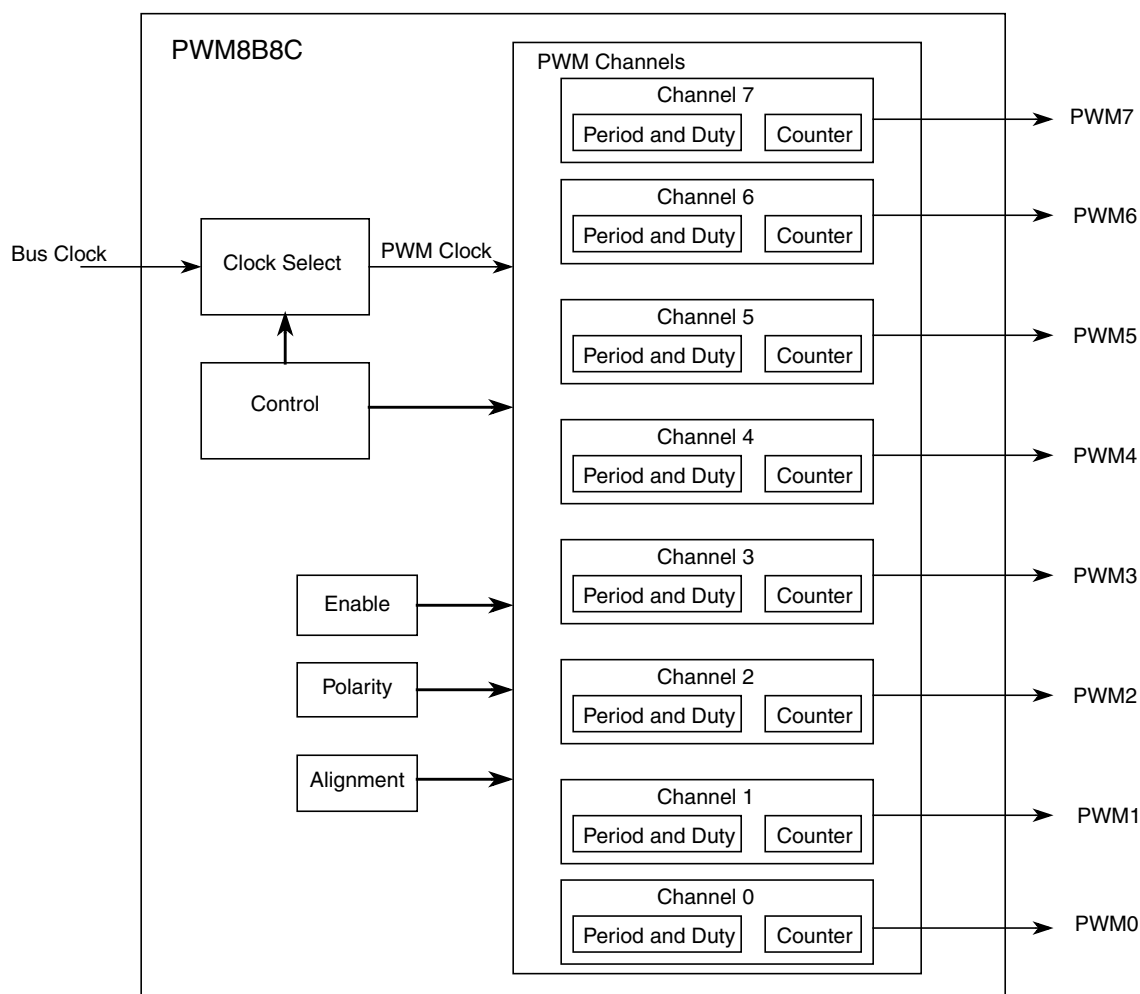


Figure 14-1. PWM Block Diagram

14.2 External Signal Description

The PWM module has a total of 8 external pins.

14.2.1 PWM7 — PWM Channel 7

This pin serves as waveform output of PWM channel 7 and as an input for the emergency shutdown feature.

14.2.2 PWM6 — PWM Channel 6

This pin serves as waveform output of PWM channel 6.

14.2.3 PWM5 — PWM Channel 5

This pin serves as waveform output of PWM channel 5.

14.2.4 PWM4 — PWM Channel 4

This pin serves as waveform output of PWM channel 4.

14.2.5 PWM3 — PWM Channel 3

This pin serves as waveform output of PWM channel 3.

14.2.6 PWM3 — PWM Channel 2

This pin serves as waveform output of PWM channel 2.

14.2.7 PWM3 — PWM Channel 1

This pin serves as waveform output of PWM channel 1.

14.2.8 PWM3 — PWM Channel 0

This pin serves as waveform output of PWM channel 0.

14.3 Memory Map and Register Definition

This section describes in detail all the registers and register bits in the PWM module.

The special-purpose registers and register bit functions that are not normally available to device end users, such as factory test control registers and reserved registers, are clearly identified by means of shading the appropriate portions of address maps and register diagrams. Notes explaining the reasons for restricting access to the registers and functions are also explained in the individual register descriptions.

14.3.1 Module Memory Map

This section describes the content of the registers in the PWM module. The base address of the PWM module is determined at the MCU level when the MCU is defined. The register decode map is fixed and begins at the first address of the module address offset. The figure below shows the registers associated with the PWM and their relative offset from the base address. The register detail description follows the order they appear in the register map.

Reserved bits within a register will always read as 0 and the write will be unimplemented. Unimplemented functions are indicated by shading the bit. .



NOTE

Register Address = Base Address + Address Offset, where the Base Address is defined at the MCU level and the Address Offset is defined at the module level.

14.3.2 Register Descriptions

This section describes in detail all the registers and register bits in the PWM module.

Register Name		Bit 7	6	5	4	3	2	1	Bit 0
0x0000 PWME	R W	PWME7	PWME6	PWME5	PWME4	PWME3	PWME2	PWME1	PWME0
0x0001 PWMPOL	R W	PPOL7	PPOL6	PPOL5	PPOL4	PPOL3	PPOL2	PPOL1	PPOL0
0x0002 PWMCLK	R W	PCLK7	PCLK6	PCLK5	PCLK4	PCLK3	PCLK2	PCLK1	PCLK0
0x0003 PWMPRCLK	R W	0	PCKB2	PCKB1	PCKB0	0	PCKA2	PCKA1	PCKA0
0x0004 PWMCAP	R W	CAE7	CAE6	CAE5	CAE4	CAE3	CAE2	CAE1	CAE0
0x0005 PWMCTL	R W	CON67	CON45	CON23	CON01	PSWAI	PFRZ	0	0
0x0006 PWMTST ¹	R W	0	0	0	0	0	0	0	0
0x0007 PWMPRSC ¹	R W	0	0	0	0	0	0	0	0
0x0008 PWMSCLA	R W	Bit 7	6	5	4	3	2	1	Bit 0
0x0009 PWMSCLB	R W	Bit 7	6	5	4	3	2	1	Bit 0
0x000A PWMSCNTA ¹	R W	0	0	0	0	0	0	0	0

 = Unimplemented or Reserved

Figure 14-2. PWM Register Summary (Sheet 1 of 3)

Register Name		Bit 7	6	5	4	3	2	1	Bit 0
0x000B PWMSCNTB ₁	R	0	0	0	0	0	0	0	0
	W								
0x000C PWMCNT0	R	Bit 7	6	5	4	3	2	1	Bit 0
	W	0	0	0	0	0	0	0	0
0x000D PWMCNT1	R	Bit 7	6	5	4	3	2	1	Bit 0
	W	0	0	0	0	0	0	0	0
0x000E PWMCNT2	R	Bit 7	6	5	4	3	2	1	Bit 0
	W	0	0	0	0	0	0	0	0
0x000F PWMCNT3	R	Bit 7	6	5	4	3	2	1	Bit 0
	W	0	0	0	0	0	0	0	0
0x0010 PWMCNT4	R	Bit 7	6	5	4	3	2	1	Bit 0
	W	0	0	0	0	0	0	0	0
0x0011 PWMCNT5	R	Bit 7	6	5	4	3	2	1	Bit 0
	W	0	0	0	0	0	0	0	0
0x0012 PWMCNT6	R	Bit 7	6	5	4	3	2	1	Bit 0
	W	0	0	0	0	0	0	0	0
0x0013 PWMCNT7	R	Bit 7	6	5	4	3	2	1	Bit 0
	W	0	0	0	0	0	0	0	0
0x0014 PWMPER0	R	Bit 7	6	5	4	3	2	1	Bit 0
	W								
0x0015 PWMPER1	R	Bit 7	6	5	4	3	2	1	Bit 0
	W								
0x0016 PWMPER2	R	Bit 7	6	5	4	3	2	1	Bit 0
	W								
0x0017 PWMPER3	R	Bit 7	6	5	4	3	2	1	Bit 0
	W								
0x0018 PWMPER4	R	Bit 7	6	5	4	3	2	1	Bit 0
	W								
0x0019 PWMPER5	R	Bit 7	6	5	4	3	2	1	Bit 0
	W								

= Unimplemented or Reserved

Figure 14-2. PWM Register Summary (Sheet 2 of 3)

Register Name		Bit 7	6	5	4	3	2	1	Bit 0
0x001A PWMPER6	R W	Bit 7	6	5	4	3	2	1	Bit 0
0x001B PWMPER7	R W	Bit 7	6	5	4	3	2	1	Bit 0
0x001C PWMDTY0	R W	Bit 7	6	5	4	3	2	1	Bit 0
0x001D PWMDTY1	R W	Bit 7	6	5	4	3	2	1	Bit 0
0x001E PWMDTY2	R W	Bit 7	6	5	4	3	2	1	Bit 0
0x001F PWMDTY3	R W	Bit 7	6	5	4	3	2	1	Bit 0
0x0010 PWMDTY4	R W	Bit 7	6	5	4	3	2	1	Bit 0
0x0021 PWMDTY5	R W	Bit 7	6	5	4	3	2	1	Bit 0
0x0022 PWMDTY6	R W	Bit 7	6	5	4	3	2	1	Bit 0
0x0023 PWMDTY7	R W	Bit 7	6	5	4	3	2	1	Bit 0
0x0024 PWMSDN	R W	PWMIF	PWMIE	0 PWMRSTRT	PWMLVL	0	PWM7IN	PWM7INL	PWM7ENA

 = Unimplemented or Reserved

Figure 14-2. PWM Register Summary (Sheet 3 of 3)

¹ Intended for factory test purposes only.

14.3.2.1 PWM Enable Register (PWME)

Each PWM channel has an enable bit (PWME_x) to start its waveform output. When any of the PWME_x bits are set (PWME_x = 1), the associated PWM output is enabled immediately. However, the actual PWM waveform is not available on the associated PWM output until its clock source begins its next cycle due to the synchronization of PWME_x and the clock source.

NOTE

The first PWM cycle after enabling the channel can be irregular.

An exception to this is when channels are concatenated. Once concatenated mode is enabled (CONxx bits set in PWMCTL register), enabling/disabling the corresponding 16-bit PWM channel is controlled by the low order PWME_x bit. In this case, the high order bytes PWME_x bits have no effect and their corresponding PWM output lines are disabled.

While in run mode, if all eight PWM channels are disabled (PWME7–0 = 0), the prescaler counter shuts off for power savings.

Module Base + 0x0000

	7	6	5	4	3	2	1	0
R	PWME7	PWME6	PWME5	PWME4	PWME3	PWME2	PWME1	PWME0
W								
Reset	0	0	0	0	0	0	0	0

Figure 14-3. PWM Enable Register (PWME)

Read: Anytime

Write: Anytime

Table 14-1. PWME Field Descriptions

Field	Description
7 PWME7	Pulse Width Channel 7 Enable 0 Pulse width channel 7 is disabled. 1 Pulse width channel 7 is enabled. The pulse modulated signal becomes available at PWM output bit 7 when its clock source begins its next cycle.
6 PWME6	Pulse Width Channel 6 Enable 0 Pulse width channel 6 is disabled. 1 Pulse width channel 6 is enabled. The pulse modulated signal becomes available at PWM output bit6 when its clock source begins its next cycle. If CON67=1, then bit has no effect and PWM output line 6 is disabled.
5 PWME5	Pulse Width Channel 5 Enable 0 Pulse width channel 5 is disabled. 1 Pulse width channel 5 is enabled. The pulse modulated signal becomes available at PWM output bit 5 when its clock source begins its next cycle.
4 PWME4	Pulse Width Channel 4 Enable 0 Pulse width channel 4 is disabled. 1 Pulse width channel 4 is enabled. The pulse modulated signal becomes available at PWM, output bit 4 when its clock source begins its next cycle. If CON45 = 1, then bit has no effect and PWM output bit4 is disabled.
3 PWME3	Pulse Width Channel 3 Enable 0 Pulse width channel 3 is disabled. 1 Pulse width channel 3 is enabled. The pulse modulated signal becomes available at PWM, output bit 3 when its clock source begins its next cycle.
2 PWME2	Pulse Width Channel 2 Enable 0 Pulse width channel 2 is disabled. 1 Pulse width channel 2 is enabled. The pulse modulated signal becomes available at PWM, output bit 2 when its clock source begins its next cycle. If CON23 = 1, then bit has no effect and PWM output bit2 is disabled.

Table 14-1. PWME Field Descriptions (continued)

Field	Description
1 PWME1	Pulse Width Channel 1 Enable 0 Pulse width channel 1 is disabled. 1 Pulse width channel 1 is enabled. The pulse modulated signal becomes available at PWM, output bit 1 when its clock source begins its next cycle.
0 PWME0	Pulse Width Channel 0 Enable 0 Pulse width channel 0 is disabled. 1 Pulse width channel 0 is enabled. The pulse modulated signal becomes available at PWM, output bit 0 when its clock source begins its next cycle. If CON01 = 1, then bit has no effect and PWM output line0 is disabled.

14.3.2.2 PWM Polarity Register (PWMPOL)

The starting polarity of each PWM channel waveform is determined by the associated PPOLx bit in the PWMPOL register. If the polarity bit is one, the PWM channel output is high at the beginning of the cycle and then goes low when the duty count is reached. Conversely, if the polarity bit is zero, the output starts low and then goes high when the duty count is reached.

Module Base + 0x0001

	7	6	5	4	3	2	1	0
R	PPOL7	PPOL6	PPOL5	PPOL4	PPOL3	PPOL2	PPOL1	PPOL0
W								
Reset	0	0	0	0	0	0	0	0

Figure 14-4. PWM Polarity Register (PWMPOL)

Read: Anytime

Write: Anytime

NOTE

PPOLx register bits can be written anytime. If the polarity is changed while a PWM signal is being generated, a truncated or stretched pulse can occur during the transition

Table 14-2. PWMPOL Field Descriptions

Field	Description
7–0 PPOL[7:0]	Pulse Width Channel 7–0 Polarity Bits 0 PWM channel 7–0 outputs are low at the beginning of the period, then go high when the duty count is reached. 1 PWM channel 7–0 outputs are high at the beginning of the period, then go low when the duty count is reached.

14.3.2.3 PWM Clock Select Register (PWMCLK)

Each PWM channel has a choice of two clocks to use as the clock source for that channel as described below.

Module Base + 0x0002

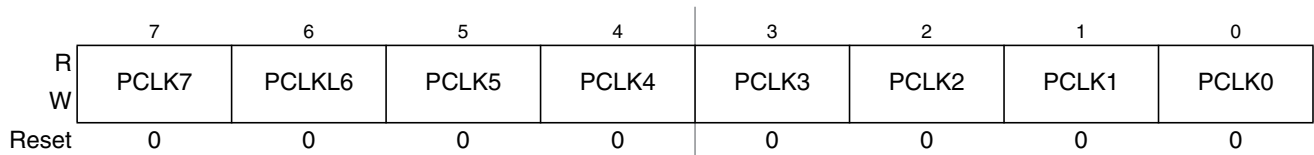


Figure 14-5. PWM Clock Select Register (PWMCLK)

Read: Anytime

Write: Anytime

NOTE

Register bits PCLK0 to PCLK7 can be written anytime. If a clock select is changed while a PWM signal is being generated, a truncated or stretched pulse can occur during the transition.

Table 14-3. PWMCLK Field Descriptions

Field	Description
7 PCLK7	Pulse Width Channel 7 Clock Select 0 Clock B is the clock source for PWM channel 7. 1 Clock SB is the clock source for PWM channel 7.
6 PCLK6	Pulse Width Channel 6 Clock Select 0 Clock B is the clock source for PWM channel 6. 1 Clock SB is the clock source for PWM channel 6.
5 PCLK5	Pulse Width Channel 5 Clock Select 0 Clock A is the clock source for PWM channel 5. 1 Clock SA is the clock source for PWM channel 5.
4 PCLK4	Pulse Width Channel 4 Clock Select 0 Clock A is the clock source for PWM channel 4. 1 Clock SA is the clock source for PWM channel 4.
3 PCLK3	Pulse Width Channel 3 Clock Select 0 Clock B is the clock source for PWM channel 3. 1 Clock SB is the clock source for PWM channel 3.
2 PCLK2	Pulse Width Channel 2 Clock Select 0 Clock B is the clock source for PWM channel 2. 1 Clock SB is the clock source for PWM channel 2.
1 PCLK1	Pulse Width Channel 1 Clock Select 0 Clock A is the clock source for PWM channel 1. 1 Clock SA is the clock source for PWM channel 1.
0 PCLK0	Pulse Width Channel 0 Clock Select 0 Clock A is the clock source for PWM channel 0. 1 Clock SA is the clock source for PWM channel 0.

14.3.2.4 PWM Prescale Clock Select Register (PWMPRCLK)

This register selects the prescale clock source for clocks A and B independently.



Module Base + 0x0003

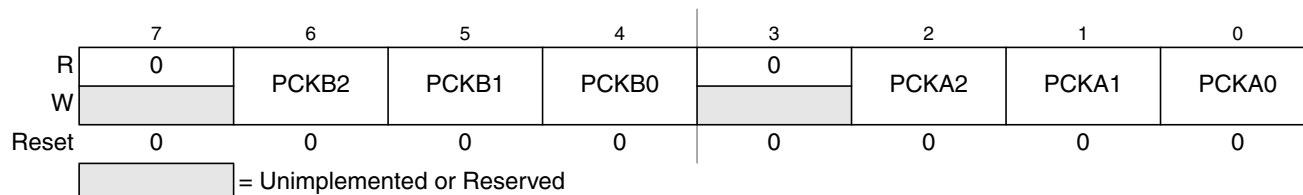


Figure 14-6. PWM Prescale Clock Select Register (PWMPRCLK)

Read: Anytime

Write: Anytime

NOTE

PCKB2–0 and PCKA2–0 register bits can be written anytime. If the clock pre-scale is changed while a PWM signal is being generated, a truncated or stretched pulse can occur during the transition.

Table 14-4. PWMPRCLK Field Descriptions

Field	Description
6–4 PCKB[2:0]	Prescaler Select for Clock B — Clock B is one of two clock sources which can be used for channels 2, 3, 6, or 7. These three bits determine the rate of clock B, as shown in Table 14-5.
2–0 PCKA[2:0]	Prescaler Select for Clock A — Clock A is one of two clock sources which can be used for channels 0, 1, 4 or 5. These three bits determine the rate of clock A, as shown in Table 14-6.

Table 14-5. Clock B Prescaler Selects

PCKB2	PCKB1	PCKB0	Value of Clock B
0	0	0	Bus clock
0	0	1	Bus clock / 2
0	1	0	Bus clock / 4
0	1	1	Bus clock / 8
1	0	0	Bus clock / 16
1	0	1	Bus clock / 32
1	1	0	Bus clock / 64
1	1	1	Bus clock / 128

Table 14-6. Clock A Prescaler Selects

PCKA2	PCKA1	PCKA0	Value of Clock A
0	0	0	Bus clock
0	0	1	Bus clock / 2
0	1	0	Bus clock / 4
0	1	1	Bus clock / 8
1	0	0	Bus clock / 16
1	0	1	Bus clock / 32
1	1	0	Bus clock / 64
1	1	1	Bus clock / 128

14.3.2.5 PWM Center Align Enable Register (PWMCAE)

The PWMCAE register contains eight control bits for the selection of center aligned outputs or left aligned outputs for each PWM channel. If the CAEx bit is set to a one, the corresponding PWM output will be center aligned. If the CAEx bit is cleared, the corresponding PWM output will be left aligned. See [Section 14.4.2.5, “Left Aligned Outputs”](#) and [Section 14.4.2.6, “Center Aligned Outputs”](#) for a more detailed description of the PWM output modes.

Module Base + 0x0004

	7	6	5	4	3	2	1	0
R	CAE7	CAE6	CAE5	CAE4	CAE3	CAE2	CAE1	CAE0
W								
Reset	0	0	0	0	0	0	0	0

Figure 14-7. PWM Center Align Enable Register (PWMCAE)

Read: Anytime

Write: Anytime

NOTE

Write these bits only when the corresponding channel is disabled.

Table 14-7. PWMCAE Field Descriptions

Field	Description
7–0 CAE[7:0]	Center Aligned Output Modes on Channels 7–0 0 Channels 7–0 operate in left aligned output mode. 1 Channels 7–0 operate in center aligned output mode.

14.3.2.6 PWM Control Register (PWMCTL)

The PWMCTL register provides for various control of the PWM module.

Module Base + 0x0005

	7	6	5	4	3	2	1	0
R	CON67	CON45	CON23	CON01	PSWAI	PFRZ	0	0
W								
Reset	0	0	0	0	0	0	0	0

 = Unimplemented or Reserved

Figure 14-8. PWM Control Register (PWMCTL)

Read: Anytime

Write: Anytime

There are three control bits for concatenation, each of which is used to concatenate a pair of PWM channels into one 16-bit channel. When channels 6 and 7 are concatenated, channel 6 registers become the high order bytes of the double byte channel. When channels 4 and 5 are concatenated, channel 4 registers become the high order bytes of the double byte channel. When channels 2 and 3 are concatenated, channel

2 registers become the high order bytes of the double byte channel. When channels 0 and 1 are concatenated, channel 0 registers become the high order bytes of the double byte channel.

See Section 14.4.2.7, “PWM 16-Bit Functions” for a more detailed description of the concatenation PWM Function.

NOTE

Change these bits only when both corresponding channels are disabled.

Table 14-8. PWMCTL Field Descriptions

Field	Description
7 CON67	Concatenate Channels 6 and 7 0 Channels 6 and 7 are separate 8-bit PWMs. 1 Channels 6 and 7 are concatenated to create one 16-bit PWM channel. Channel 6 becomes the high order byte and channel 7 becomes the low order byte. Channel 7 output pin is used as the output for this 16-bit PWM (bit 7 of port PWMP). Channel 7 clock select control-bit determines the clock source, channel 7 polarity bit determines the polarity, channel 7 enable bit enables the output and channel 7 center aligned enable bit determines the output mode.
6 CON45	Concatenate Channels 4 and 5 0 Channels 4 and 5 are separate 8-bit PWMs. 1 Channels 4 and 5 are concatenated to create one 16-bit PWM channel. Channel 4 becomes the high order byte and channel 5 becomes the low order byte. Channel 5 output pin is used as the output for this 16-bit PWM (bit 5 of port PWMP). Channel 5 clock select control-bit determines the clock source, channel 5 polarity bit determines the polarity, channel 5 enable bit enables the output and channel 5 center aligned enable bit determines the output mode.
5 CON23	Concatenate Channels 2 and 3 0 Channels 2 and 3 are separate 8-bit PWMs. 1 Channels 2 and 3 are concatenated to create one 16-bit PWM channel. Channel 2 becomes the high order byte and channel 3 becomes the low order byte. Channel 3 output pin is used as the output for this 16-bit PWM (bit 3 of port PWMP). Channel 3 clock select control-bit determines the clock source, channel 3 polarity bit determines the polarity, channel 3 enable bit enables the output and channel 3 center aligned enable bit determines the output mode.
4 CON01	Concatenate Channels 0 and 1 0 Channels 0 and 1 are separate 8-bit PWMs. 1 Channels 0 and 1 are concatenated to create one 16-bit PWM channel. Channel 0 becomes the high order byte and channel 1 becomes the low order byte. Channel 1 output pin is used as the output for this 16-bit PWM (bit 1 of port PWMP). Channel 1 clock select control-bit determines the clock source, channel 1 polarity bit determines the polarity, channel 1 enable bit enables the output and channel 1 center aligned enable bit determines the output mode.
3 PSWAI	PWM Stops in Wait Mode — Enabling this bit allows for lower power consumption in wait mode by disabling the input clock to the prescaler. 0 Allow the clock to the prescaler to continue while in wait mode. 1 Stop the input clock to the prescaler whenever the MCU is in wait mode.
2 PFRZ	PWM Counters Stop in Freeze Mode — In freeze mode, there is an option to disable the input clock to the prescaler by setting the PFRZ bit in the PWMCTL register. If this bit is set, whenever the MCU is in freeze mode, the input clock to the prescaler is disabled. This feature is useful during emulation as it allows the PWM function to be suspended. In this way, the counters of the PWM can be stopped while in freeze mode so that once normal program flow is continued, the counters are re-enabled to simulate real-time operations. Since the registers can still be accessed in this mode, to re-enable the prescaler clock, either disable the PFRZ bit or exit freeze mode. 0 Allow PWM to continue while in freeze mode. 1 Disable PWM input clock to the prescaler whenever the part is in freeze mode. This is useful for emulation.

14.3.2.7 Reserved Register (PWMTST)

This register is reserved for factory testing of the PWM module and is not available in normal modes.

Module Base + 0x0006

	7	6	5	4	3	2	1	0
R	0	0	0	0	0	0	0	0
W								
Reset	0	0	0	0	0	0	0	0

= Unimplemented or Reserved

Figure 14-9. Reserved Register (PWMTST)

Read: Always read \$00 in normal modes

Write: Unimplemented in normal modes

NOTE

Writing to this register when in special modes can alter the PWM functionality.

14.3.2.8 Reserved Register (PWMPRSC)

This register is reserved for factory testing of the PWM module and is not available in normal modes.

Module Base + 0x0007

	7	6	5	4	3	2	1	0
R	0	0	0	0	0	0	0	0
W								
Reset	0	0	0	0	0	0	0	0

= Unimplemented or Reserved

Figure 14-10. Reserved Register (PWMPRSC)

Read: Always read \$00 in normal modes

Write: Unimplemented in normal modes

NOTE

Writing to this register when in special modes can alter the PWM functionality.

14.3.2.9 PWM Scale A Register (PWMSCLA)

PWMSCLA is the programmable scale value used in scaling clock A to generate clock SA. Clock SA is generated by taking clock A, dividing it by the value in the PWMSCLA register and dividing that by two.

$$\text{Clock SA} = \text{Clock A} / (2 * \text{PWMSCLA})$$

NOTE

When PWMSCLA = \$00, PWMSCLA value is considered a full scale value of 256. Clock A is thus divided by 512.

Any value written to this register will cause the scale counter to load the new scale value (PWMSCLA).

Module Base + 0x0008

	7	6	5	4	3	2	1	0
R								
W	Bit 7	6	5	4	3	2	1	Bit 0
Reset	0	0	0	0	0	0	0	0

Figure 14-11. PWM Scale A Register (PWMSCLA)

Read: Anytime

Write: Anytime (causes the scale counter to load the PWMSCLA value)

14.3.2.10 PWM Scale B Register (PWMSCLB)

PWMSCLB is the programmable scale value used in scaling clock B to generate clock SB. Clock SB is generated by taking clock B, dividing it by the value in the PWMSCLB register and dividing that by two.

$$\text{Clock SB} = \text{Clock B} / (2 * \text{PWMSCLB})$$

NOTE

When PWMSCLB = \$00, PWMSCLB value is considered a full scale value of 256. Clock B is thus divided by 512.

Any value written to this register will cause the scale counter to load the new scale value (PWMSCLB).

Module Base + 0x0009

	7	6	5	4	3	2	1	0
R								
W	Bit 7	6	5	4	3	2	1	Bit 0
Reset	0	0	0	0	0	0	0	0

Figure 14-12. PWM Scale B Register (PWMSCLB)

Read: Anytime

Write: Anytime (causes the scale counter to load the PWMSCLB value).

14.3.2.11 Reserved Registers (PWMSCNTx)

The registers PWMSCNTA and PWMSCNTB are reserved for factory testing of the PWM module and are not available in normal modes.

Module Base + 0x000A, 0x000B

	7	6	5	4	3	2	1	0
R	0	0	0	0	0	0	0	0
W								
Reset	0	0	0	0	0	0	0	0

= Unimplemented or Reserved

Figure 14-13. Reserved Registers (PWMSCNTx)

Read: Always read \$00 in normal modes

Write: Unimplemented in normal modes

NOTE

Writing to these registers when in special modes can alter the PWM functionality.

14.3.2.12 PWM Channel Counter Registers (PWMCNTx)

Each channel has a dedicated 8-bit up/down counter which runs at the rate of the selected clock source. The counter can be read at any time without affecting the count or the operation of the PWM channel. In left aligned output mode, the counter counts from 0 to the value in the period register - 1. In center aligned output mode, the counter counts from 0 up to the value in the period register and then back down to 0.

Any value written to the counter causes the counter to reset to \$00, the counter direction to be set to up, the immediate load of both duty and period registers with values from the buffers, and the output to change according to the polarity bit. The counter is also cleared at the end of the effective period (see [Section 14.4.2.5, “Left Aligned Outputs”](#) and [Section 14.4.2.6, “Center Aligned Outputs”](#) for more details). When the channel is disabled (PWME_x = 0), the PWMCNT_x register does not count. When a channel becomes enabled (PWME_x = 1), the associated PWM counter starts at the count in the PWMCNT_x register. For more detailed information on the operation of the counters, see [Section 14.4.2.4, “PWM Timer Counters”](#).

In concatenated mode, writes to the 16-bit counter by using a 16-bit access or writes to either the low or high order byte of the counter will reset the 16-bit counter. Reads of the 16-bit counter must be made by 16-bit access to maintain data coherency.

NOTE

Writing to the counter while the channel is enabled can cause an irregular PWM cycle to occur.

Module Base + 0x000C = PWMCNT0, 0x000D = PWMCNT1, 0x000E = PWMCNT2, 0x000F = PWMCNT3

Module Base + 0x0010 = PWMCNT4, 0x0011 = PWMCNT5, 0x0012 = PWMCNT6, 0x0013 = PWMCNT7

	7	6	5	4	3	2	1	0
R	Bit 7	6	5	4	3	2	1	Bit 0
W	0	0	0	0	0	0	0	0
Reset	0	0	0	0	0	0	0	0

Figure 14-14. PWM Channel Counter Registers (PWMCNTx)

Read: Anytime

Write: Anytime (any value written causes PWM counter to be reset to \$00).

14.3.2.13 PWM Channel Period Registers (PWMPERx)

There is a dedicated period register for each channel. The value in this register determines the period of the associated PWM channel.

The period registers for each channel are double buffered so that if they change while the channel is enabled, the change will NOT take effect until one of the following occurs:

- The effective period ends
- The counter is written (counter resets to \$00)
- The channel is disabled

In this way, the output of the PWM will always be either the old waveform or the new waveform, not some variation in between. If the channel is not enabled, then writes to the period register will go directly to the latches as well as the buffer.

NOTE

Reads of this register return the most recent value written. Reads do not necessarily return the value of the currently active period due to the double buffering scheme.

See [Section 14.4.2.3, “PWM Period and Duty”](#) for more information.

To calculate the output period, take the selected clock source period for the channel of interest (A, B, SA, or SB) and multiply it by the value in the period register for that channel:

- Left aligned output (CAEx = 0)
- $\text{PWMx Period} = \text{Channel Clock Period} * \text{PWMPERx Center Aligned Output (CAEx = 1)}$

$$\text{PWMx Period} = \text{Channel Clock Period} * (2 * \text{PWMPERx})$$

For boundary case programming values, please refer to [Section 14.4.2.8, “PWM Boundary Cases”](#).

Module Base + 0x0014 = PWMPER0, 0x0015 = PWMPER1, 0x0016 = PWMPER2, 0x0017 = PWMPER3

Module Base + 0x0018 = PWMPER4, 0x0019 = PWMPER5, 0x001A = PWMPER6, 0x001B = PWMPER7

	7	6	5	4	3	2	1	0
R	Bit 7	6	5	4	3	2	1	Bit 0
W								
Reset	1	1	1	1	1	1	1	1

Figure 14-15. PWM Channel Period Registers (PWMPERx)

Read: Anytime

Write: Anytime

14.3.2.14 PWM Channel Duty Registers (PWMDTYx)

There is a dedicated duty register for each channel. The value in this register determines the duty of the associated PWM channel. The duty value is compared to the counter and if it is equal to the counter value a match occurs and the output changes state.

The duty registers for each channel are double buffered so that if they change while the channel is enabled, the change will NOT take effect until one of the following occurs:

- The effective period ends
- The counter is written (counter resets to \$00)
- The channel is disabled

In this way, the output of the PWM will always be either the old duty waveform or the new duty waveform, not some variation in between. If the channel is not enabled, then writes to the duty register will go directly to the latches as well as the buffer.

NOTE

Reads of this register return the most recent value written. Reads do not necessarily return the value of the currently active duty due to the double buffering scheme.

See [Section 14.4.2.3, “PWM Period and Duty”](#) for more information.

NOTE

Depending on the polarity bit, the duty registers will contain the count of either the high time or the low time. If the polarity bit is one, the output starts high and then goes low when the duty count is reached, so the duty registers contain a count of the high time. If the polarity bit is zero, the output starts low and then goes high when the duty count is reached, so the duty registers contain a count of the low time.

To calculate the output duty cycle (high time as a% of period) for a particular channel:

- Polarity = 0 (PPOL x =0)

$$\text{Duty Cycle} = [(PWMPERx - PWMDTYx) / PWMPERx] * 100\%$$
- Polarity = 1 (PPOLx = 1)

$$\text{Duty Cycle} = [PWMDTYx / PWMPERx] * 100\%$$

For boundary case programming values, please refer to [Section 14.4.2.8, “PWM Boundary Cases”](#).

Module Base + 0x001C = PWMDTY0, 0x001D = PWMDTY1, 0x001E = PWMDTY2, 0x001F = PWMDTY3
 Module Base + 0x0020 = PWMDTY4, 0x0021 = PWMDTY5, 0x0022 = PWMDTY6, 0x0023 = PWMDTY7

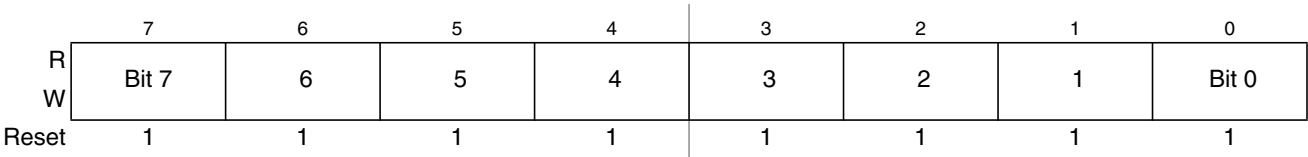


Figure 14-16. PWM Channel Duty Registers (PWMDTYx)

Read: Anytime

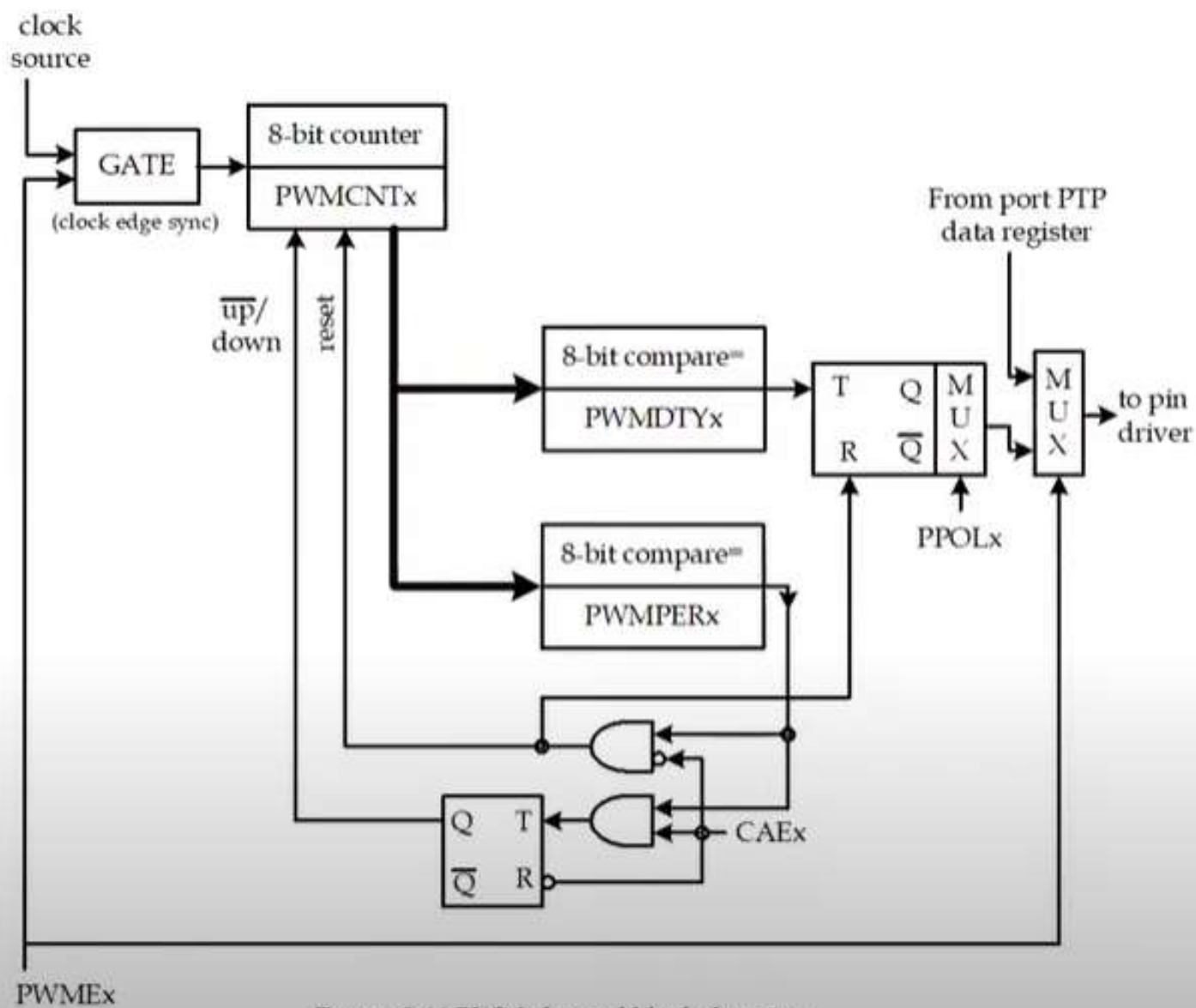


Figure 8.44 PWM channel block diagram